

This code is a Biscoff Cookie Grocery Tracker that checks whether the user already has the ingredients needed for a recipe. The program uses lists to store recipe ingredients and pantry items, loops to repeatedly gather user input, conditional statements to validate and compare ingredients, and a function to identify missing items. It also uses list methods and string processing to organize the grocery list and display a final inventory summary for the user.

```
In [1]: def check_missing_ingredients(recipe_list, pantry_list):
        missing = list()

        for ingredient in recipe_list:
            if ingredient not in pantry_list:
                missing.append(ingredient)
        return missing

biscoff_recipe = ['vanilla', 'biscoff', 'flour', 'brown sugar', 'butter', 'eggs', 'baking powder', 'chocolate_chip']

my_pantry = list()

print('Biscoff Cookie Grocery Tracker')
print('Recipe requires:', len(biscoff_recipe), 'key items.')

while True:
    print('What do you currently have in your pantry?')
    item = input('Enter ingredient name (or type "done"): ')

    item = item.strip().lower()

    if item == 'done':
        break

    if item in biscoff_recipe:
        if item not in my_pantry:
            my_pantry.append(item)
            print('Great! Added', item, 'to your pantry inventory.')
        else:
            print('You already added that item.')
    else:
        print('Note:', item, 'is fine to have, but it is not required for this Biscoff recipe.')

grocery_list = check_missing_ingredients(biscoff_recipe, my_pantry)
```

```
print('Inventory Summary')
print('Items you have:', len(my_pantry), '/', len(biscoff_recipe))

if len(grocery_list) == 0:
    print('SUCCESS! You have all ingredients. Time to bake some Biscoff cookies!')
else:
    print('YOUR OFFICIAL GROCERY LIST')
    grocery_list.sort()

    count = 1
    for missing_item in grocery_list:
        print(count, '-', missing_item)
        count = count + 1

    print('You still need to buy', len(grocery_list), 'items.')
```

Biscoff Cookie Grocery Tracker

Recipe requires: 8 key items.

What do you currently have in your pantry?

Great! Added baking powder to your pantry inventory.

What do you currently have in your pantry?

Great! Added vanilla to your pantry inventory.

What do you currently have in your pantry?

Inventory Summary

Items you have: 2 / 8

YOUR OFFICIAL GROCERY LIST

1 - biscoff

2 - brown sugar

3 - butter

4 - chocolate_chip

5 - eggs

6 - flour

You still need to buy 6 items.